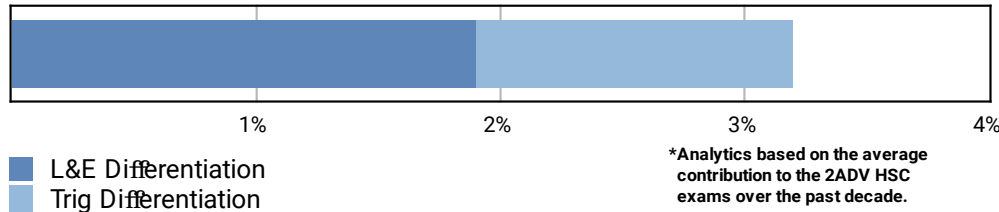


ADV: Calculus (Adv), C2 Differential Calculus (Adv)
L&E Differentiation (Y12)

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Exam Equivalent Time: 73.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C2 Trig and L&E Differentiation



HISTORICAL CONTRIBUTION

- C2 Log/Exponential and Trig Differentiation has contributed a healthy 3.2% to past Advanced exams, on average over the past decade.
- This topic has been split into two sub-topics for analysis purposes: 1-Log and Exponential Differentiation (1.9%), and 2-Trig Differentiation (1.3%).
- This analysis looks at L&E Differentiation.

HSC ANALYSIS - What to expect and common pitfalls

- This topic area is the definition of low lying fruit where well prepared students should expect to score highly although we note that stand alone questions have been absent in the 2020-21 exam papers.
- L&E Differentiation as a stand alone sub-topic was consistently examined before 2020 and offers up some of the easiest marks in the Advanced exam, typically around the band 3 difficulty level.
- HSC Marker's Comments have noted that students who wrote out the quotient rule and all its inputs before substituting in values made less mistakes.
- The new syllabus explicitly mentions the differentiation of a^x and $\log_a x$ which are both covered in the database within SM-Bank questions.
- Notwithstanding the omission in the 2020-21 papers, this topic is crucial gateway knowledge for numerous later Y12 topics and must be a core competency of every student.

Questions

1. Calculus, 2ADV C2 2016 HSC 5 MC

What is the derivative of $\ln(\cos x)$?

- (A) $-\sec x$
- (B) $-\tan x$
- (C) $\sec x$
- (D) $\tan x$

2. Calculus, 2ADV C2 2017 HSC 3 MC

What is the derivative of e^{x^2} ?

- (A) $x^2 e^{x^2-1}$
- (B) $2xe^{2x}$
- (C) $2xe^{x^2}$
- (D) $2e^{x^2}$

3. Calculus, 2ADV C4 2018 HSC 5 MC

What is the derivative of $\sin(\ln x)$, where $x > 0$?

- (A) $\cos\left(\frac{1}{x}\right)$
- (B) $\cos(\ln x)$
- (C) $\cos\left(\frac{\ln x}{x}\right)$
- (D) $\frac{\cos(\ln x)}{x}$

4. Calculus, 2ADV C2 EQ-Bank 4 MC

If $f(x) = \log_2(x^{2x})$, which expression is equal to $f'(x)$?

(A) $\frac{2}{x^{2x} \ln 2}$

(B) $\frac{2}{\ln 2} + 2 \log_2 x$

(C) $\log_2 x + \frac{2}{\ln 2}$

(D) $\frac{2}{\ln 2} \times \log_2(x^{2x-1})$

5. Calculus, 2ADV C2 SM-Bank 1 MC

If $f(x) = e^{g(x^2)}$, where g is a differentiable function, then $f'(x)$ is equal to

A. $2xe^{g(x^2)}$

B. $2xg(x^2)e^{g(x^2)}$

C. $2xg'(x^2)e^{g(x^2)}$

D. $2xg'(2x)e^{g(x^2)}$

6. Calculus, 2ADV C2 2008 HSC 2aii

Differentiate with respect to x :

$x^2 \log_e x$ (2 marks)

7. Calculus, 2ADV C2 SM-Bank 13

Differentiate $y = 2e^{-3x}$ with respect to x . (2 mark)

8. Calculus, 2ADV C2 SM-Bank 7

Let $f(x) = \frac{e^x}{(x^2 - 3)}$.

Find $f'(x)$. (2 marks)

9. Calculus, 2ADV C2 SM-Bank 8

Let $y = (x + 5)\log_e(x)$.

Find $\frac{dy}{dx}$ when $x = 5$. (2 marks)

10. Calculus, 2ADV C2 2019 MET1 1a

Let $y = \frac{2e^{2x} - 1}{e^x}$.

Find $\frac{dy}{dx}$. (2 marks)

11. Calculus, 2ADV C3 2005 HSC 2d

Find the equation of the tangent to $y = \log_e x$ at the point $(e, 1)$. (2 marks)

12. Calculus, 2ADV C2 2009 HSC 2aii

Differentiate with respect to x .

$(e^x + 1)^2$. (2 marks)

13. Calculus, 2ADV C2 2011 HSC 1d

Differentiate $\ln(5x + 2)$ with respect to x . (2 marks)

14. Calculus, 2ADV C2 2012 HSC 11d

Differentiate $(3 + e^{2x})^5$. (2 marks)

15. Calculus, 2ADV C2 2012 HSC 12ai

Differentiate with respect to x

$(x - 1)\log_e x$ (2 marks)

16. Calculus, 2ADV C2 2013 HSC 11d

Differentiate $x^2 e^x$ (2 marks)

17. Calculus, 2ADV C2 2007 HSC 2ai

Differentiate with respect to x :

$\frac{2x}{e^x + 1}$ (2 marks)

18. Calculus, 2ADV C2 2015 HSC 11e

Differentiate $(e^x + x)^5$. (2 marks)

19. Calculus, 2ADV C2 2015 HSC 11f

Differentiate $y = (x + 4)\ln x$. (2 marks)

20. Calculus, 2ADV C2 2017 HSC 11d

Differentiate $x^3 \ln x$. (2 marks)

21. Calculus, 2ADV C2 2018 HSC 11g

Differentiate $\frac{e^x}{x+1}$. (2 marks)

22. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$$e^{\tan(2x)} \quad (2 \text{ marks})$$

23. Calculus, 2ADV C2 EQ-Bank 1

Differentiate $\log_2 x^2$ with respect to x . (2 marks)

24. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$$10^{5x^2-3x} \quad (2 \text{ marks})$$

25. Calculus, 2ADV C2 EQ-Bank 3

Differentiate $3x 6^x$. (2 marks)

26. Calculus, 2ADV C2 SM-Bank 4

Differentiate $5^{x^2} 5x$. (2 marks)

27. Calculus, 2ADV C2 SM-Bank 6

Differentiate with respect to x :

$$\log_e x^x \quad (2 \text{ marks})$$

Worked Solutions

1. Calculus, 2ADV C2 2016 HSC 5 MC

$$y = \ln(\cos x)$$

$$\frac{dy}{dx} = \frac{-\sin x}{\cos x}$$

$$= -\tan x$$

$$\Rightarrow B$$

2. Calculus, 2ADV C2 2017 HSC 3 MC

$$y = e^{x^2}$$

$$\frac{dy}{dx} = 2x e^{x^2}$$

$$\Rightarrow C$$

3. Calculus, 2ADV C4 2018 HSC 5 MC

$$y = \sin(\ln x)$$

$$\frac{dy}{dx} = \cos(\ln x) \times \frac{d}{dx}(\ln x)$$

$$= \cos(\ln x) \times \frac{1}{x}$$

$$= \frac{\cos(\ln x)}{x}$$

$$\Rightarrow D$$

4. Calculus, 2ADV C2 EQ-Bank 4 MC

$$\begin{aligned}f(x) &= \log_2(x^{2x}) \\&= 2x \log_2 x \\&= \frac{2x \ln x}{\ln 2}\end{aligned}$$

$$\begin{aligned}f'(x) &= \frac{1}{\ln 2} \left(2x \cdot \frac{1}{x} + 2 \ln x \right) \\&= \frac{2}{\ln 2} + \frac{2 \ln x}{\ln 2} \\&= \frac{2}{\ln 2} + 2 \log_2 x\end{aligned}$$

$\Rightarrow B$

5. Calculus, 2ADV C2 SM-Bank 1 MC

$$f(x) = e^{g(x^2)}$$

Using the chain rule (twice):

$$\begin{aligned}f'(x) &= \frac{d}{dx} [g(x^2)] \cdot e^{g(x^2)} \\&= 2x \cdot g'(x^2) \cdot e^{g(x^2)}\end{aligned}$$

$\Rightarrow C$

6. Calculus, 2ADV C2 2008 HSC 2aii

$$\begin{aligned}y &= x^2 \log_e x \\ \frac{dy}{dx} &= x^2 \cdot \frac{1}{x} + 2x \cdot \log_e x \\&= x + 2x \log_e x\end{aligned}$$

7. Calculus, 2ADV C2 SM-Bank 13

$$\begin{aligned}y &= 2e^{-3x} \\ \frac{dy}{dx} &= -3 \times 2e^{-3x} \\&= -6e^{-3x}\end{aligned}$$

8. Calculus, 2ADV C2 SM-Bank 7

$$\begin{aligned}\text{Let } u &= e^x \Rightarrow u' = e^x \\ v &= (x^2 - 3) \Rightarrow v' = 2x\end{aligned}$$

$$\begin{aligned}f'(x) &= \frac{e^x(x^2 - 3) - 2xe^x}{(x^2 - 3)^2} \\&= \frac{e^x(x^2 - 2x - 3)}{(x^2 - 3)^2}\end{aligned}$$

9. Calculus, 2ADV C2 SM-Bank 8

$$\begin{aligned}\frac{dy}{dx} &= 1 \times \log_e x + (x + 5) \cdot \frac{1}{x} \\&= \log_e x + \frac{x + 5}{x}\end{aligned}$$

$$\therefore \frac{dy}{dx} \Big|_{x=5} = \log_e 5 + 2$$

10. Calculus, 2ADV C2 2019 MET1 1a

Method 1

$$y = 2e^x - e^{-x}$$

$$\frac{dy}{dx} = 2e^x + e^{-x}$$

Method 2

$$\frac{dy}{dx} = \frac{4e^{2x} \cdot e^x - (2e^{2x} - 1)e^x}{(e^x)^2}$$

$$= \frac{4e^{3x} - 2e^{3x} + e^x}{e^{2x}}$$

$$= \frac{2e^{2x} + 1}{e^x}$$

11. Calculus, 2ADV C3 2005 HSC 2d

$$y = \log_e x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

At $(e, 1)$,

$$m = \frac{1}{e}$$

Equation of tangent, $m = \frac{1}{e}$, through $(e, 1)$

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{e}(x - e)$$

$$y - 1 = \frac{x}{e} - 1$$

$$y = \frac{x}{e}$$

12. Calculus, 2ADV C2 2009 HSC 2aii

$$y = (e^x + 1)^2$$

$$\begin{aligned}\frac{dy}{dx} &= 2(e^x + 1)^1 \times \frac{d}{dx}(e^x + 1) \\ &= 2e^x(e^x + 1)\end{aligned}$$

13. Calculus, 2ADV C2 2011 HSC 1d

$$y = \ln(5x + 2)$$

$$\frac{dy}{dx} = \frac{5}{5x + 2}$$

14. Calculus, 2ADV C2 2012 HSC 11d

$$y = (3 + e^{2x})^5$$

$$\begin{aligned}\frac{dy}{dx} &= 5(3 + e^{2x})^4 \times \frac{d}{dx}(3 + e^{2x}) \\ &= 5(3 + e^{2x})^4 \times 2e^{2x} \\ &= 10e^{2x}(3 + e^{2x})^4\end{aligned}$$

15. Calculus, 2ADV C2 2012 HSC 12ai

$$y = (x - 1)\log_e x$$

$$\begin{aligned}\frac{dy}{dx} &= 1(\log_e x) + (x - 1)\frac{1}{x} \\ &= \log_e x + 1 - \frac{1}{x}\end{aligned}$$

16. Calculus, 2ADV C2 2013 HSC 11d

Using the product rule

$$\text{Let } u = x^2, \quad u' = 2x$$

$$\text{Let } v = e^x, \quad v' = e^x$$

$$\begin{aligned} \frac{d(uv)}{dx} &= u'v + v'u \\ &= 2xe^x + x^2e^x \\ &= xe^x(x+2) \end{aligned}$$

17. Calculus, 2ADV C2 2007 HSC 2ai

$$y = \frac{2x}{e^x + 1}$$

$$u = 2x \quad v = e^x + 1$$

$$u' = 2 \quad v' = e^x$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{2(e^x + 1) - 2x(e^x)}{(e^x + 1)^2} \\ &= \frac{2e^x + 2 - 2x \cdot e^x}{(e^x + 1)^2} \\ &= \frac{2(e^x + 1 - xe^x)}{(e^x + 1)^2} \end{aligned}$$

18. Calculus, 2ADV C2 2015 HSC 11e

$$y = (e^x + x)^5$$

$$\begin{aligned} \frac{dy}{dx} &= 5(e^x + x)^4 \times \frac{d}{dx}(e^x + x) \\ &= 5(e^x + x)^4 \times (e^x + 1) \\ &= 5(e^x + 1)(e^x + x)^4 \end{aligned}$$

19. Calculus, 2ADV C2 2015 HSC 11f

$$y = (x + 4)\ln x$$

Using the product rule

$$\begin{aligned} \frac{dy}{dx} &= \frac{d}{dx}(x + 4) \cdot \ln x + (x + 4) \frac{d}{dx} \ln x \\ &= \ln x + (x + 4) \frac{1}{x} \\ &= \ln x + \frac{4}{x} + 1 \end{aligned}$$

20. Calculus, 2ADV C2 2017 HSC 11d

$$y = x^3 \ln x$$

Using the product rule:

$$\begin{aligned} \frac{dy}{dx} &= 3x^2 \cdot \ln x + x^3 \cdot \frac{1}{x} \\ &= x^2(3 \ln x + 1) \end{aligned}$$

21. Calculus, 2ADV C2 2018 HSC 11g

$$y = \frac{e^x}{x + 1}$$

Differentiate using quotient rule:

$$u = e^x \quad v = x + 1$$

$$u' = e^x \quad v' = 1$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{e^x(x + 1) - e^x \cdot 1}{(x + 1)^2} \\ &= \frac{xe^x}{(x + 1)^2} \end{aligned}$$

22. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned}
 y &= e^{\tan(2x)} \\
 \frac{dy}{dx} &= \frac{d}{dx} \tan(2x) \times e^{\tan(2x)} \\
 &= 2 \sec^2(2x) \cdot e^{\tan(2x)}
 \end{aligned}$$

23. Calculus, 2ADV C2 EQ-Bank 1

$$\begin{aligned}
 y &= \log_2 x^2 \\
 \frac{dy}{dx} &= \frac{d}{dx} \left(\frac{\ln x^2}{\ln 2} \right) \\
 &= \frac{1}{\ln 2} \cdot \frac{d}{dx} (\ln x^2) \\
 &= \frac{1}{\ln 2} \cdot \frac{2x}{x^2} \\
 &= \frac{2}{x \ln 2}
 \end{aligned}$$

TIP: The new Advanced reference sheet can be used here!

24. Calculus, 2ADV C2 EQ-Bank 2

$$\begin{aligned}
 y &= 10^{5x^2-3x} \\
 \frac{dy}{dx} &= \ln 10 (10x - 3) \cdot 10^{5x^2-3x}
 \end{aligned}$$

TIP: The new Advanced reference sheet can be used here!

25. Calculus, 2ADV C2 EQ-Bank 3

$$\begin{aligned}
 y &= 3x \cdot 6^x \\
 \frac{dy}{dx} &= 3 \cdot 6^x + \ln 6 \cdot 6^x \cdot 3x \\
 &= 3 \cdot 6^x (1 + x \ln 6)
 \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 6^x .

26. Calculus, 2ADV C2 SM-Bank 4

$$\begin{aligned}
 y &= 5^{x^2} \cdot 5x \\
 \frac{dy}{dx} &= \ln 5 \cdot 2x \cdot 5^{x^2} \cdot 5x + 5^{x^2} \cdot 5 \\
 &= 5^{x^2} (\ln 5 \cdot 10x^2 + 5) \\
 &= 5^{x^2+1} (\ln 5 \cdot 2x^2 + 1)
 \end{aligned}$$

COMMENT: See HSC exam reference sheet when differentiating 5^x .

27. Calculus, 2ADV C2 SM-Bank 6

$$\begin{aligned}
 y &= \log_e x^x \\
 &= x \log_e x \\
 \frac{dy}{dx} &= x \cdot \frac{1}{x} + \log_e x \\
 &= 1 + \log_e x
 \end{aligned}$$